Ground-to-Grid Scale Factor in the GoSurvey Traverse Editor

Proof of Calculations

REQ-401 GitHub Issue #828 PR #833

1 What problem this fixes

A field survey measures **ground distances** — the actual tape/EDM distance along the earth's curved, elevated surface. A map projection (e.g. MGA56 / GDA2020) represents the earth on a flat **grid**, which necessarily stretches or shrinks distances by a small, position-dependent factor k. A traverse closed and reported in ground distances will not agree with coordinates taken from the grid unless every distance is first reduced by k. GoSurvey's Traverse Editor computed only ground distances; this fix adds an option to reduce them to grid before they drive coordinates, closure, and the least-squares adjustment.

2 The governing formulas

GoSurvey already computes the projection's point scale factor k_{grid} (CS-MAP) and an elevation / sea-level factor $k_{\text{sea}} = R/(R+h)$ for the Drawing Settings Transformation tab (REQ-360), where R is the ellipsoid (or typed) radius and h is the elevation above it. REQ-401 reuses that exact pair — it does not introduce a second calculation:

$$k_{\text{combined}} = k_{\text{grid}} \cdot k_{\text{sea}} \tag{1}$$

$$k_{\text{grid-only}} = k_{\text{grid}} \tag{2}$$

For each traverse leg, the slope distance is first reduced to a horizontal (ground) distance d_{ground} exactly as before (REQ-012). REQ-401 inserts one multiplication before that distance is used for anything else:

$$d_{\text{grid}} = d_{\text{ground}} \cdot k \qquad \big(k = k_{\text{combined}} \text{ or } k_{\text{grid-only}},\ \text{user's choice; } k = 1 \text{ when the option is off}\big) \tag{3}$$

$$\Delta E = d_{\text{grid}} \cdot \sin(\text{bearing}) \tag{4}$$

$$\Delta N = d_{\text{grid}} \cdot \cos(\text{bearing}) \tag{5}$$

$$\text{perimeter} = \sum d_{\text{grid}} \qquad \text{(fed to closure, and to the least-squares adjustment, REQ-015)} \tag{6}$$

Source: `src/traverse/TraverseCalc.cpp`, function `ComputeTraverse` (lines ∼391–406); multiplier resolved by `src/commands/CadCommands_Geo.cpp`, function `ResolveTraverseScaleFactor`.

3 Worked example A — a single leg under a known factor

A representative central-meridian UTM-style grid factor, $k = 0.9996$, applied to a due-east (bearing 90°) leg measured at 1000.0000 ft on the ground:

Quantity	Formula	Value
Ground distance d_{ground}	measured	1000.0000 ft
Scale factor k	grid-only or combined (user choice)	0.9996
Grid distance d_{grid}	$d_{\text{ground}} \cdot k$	999.6000 ft
ΔE	$d_{\text{grid}} \cdot \sin(90°)$	999.6000 ft
ΔN	$d_{\text{grid}} \cdot \cos(90°)$	0.0000 ft
Closure perimeter (this leg)	$\sum d_{\text{grid}}$	999.6000 ft

Verified by: `tests/TraverseTests.cpp`, test case "Scale factor on scales distance, coordinates, and perimeter" — asserts `computedHorizDist`, `computedEasting` and `closurePerimeter` all equal 999.6 for this exact input, and that `computedGroundHorizDist` stays 1000.0 (the unscaled value is preserved for display). A sibling test with the option left off (the default) proves the output is unchanged from before this feature existed.

4 Worked example B — a real projection, checked against an NGS datasheet

This example proves GoSurvey's own k_{grid} at a real point, not an assumed one. The point is NGS datasheet PID AG9976 ("UNIV OF TEXAS TOWER SEC", Travis County, TX), State Plane Texas Central zone. The datasheet publishes its own computed scale factor for this exact station:

> NGS AG9976 — SPC TX-C: $E = 10{,}077{,}391.26$ ftUS, $N = 3{,}115{,}243.14$ ftUS
> published scale factor = 0.99995905

GoSurvey resolves this coordinate through its CS-MAP-backed projection engine (`geo::GridScaleFactor`) and through `ResolveTraverseScaleFactor` (the function the Traverse Editor calls) and obtains, independently:

Source	$k_{\textbf{grid-only}}$	Agreement
NGS datasheet (published)	0.99995905	— reference
`geo::GridScaleFactor` (direct CS-MAP call)	matches to $\leq 1\times10^{-8}$	within 10^{-8}
`ResolveTraverseScaleFactor`, GridOnly	0.9999590531	within 10^{-7}
`ResolveTraverseScaleFactor`, Combined (sea level off)	0.9999590531	identical to GridOnly

Sea-level (elevation) reduction, Combined factor

Turning on the Transformation tab's sea level option with an elevation $h = 1000$ ft above an ellipsoid radius $R = 20{,}906{,}000$ ft (converted to meters as the US survey foot, $R \approx 6{,}372{,}161.54$ m):

$$k_{\text{sea}} = \frac{R}{R+h} = \frac{6{,}372{,}161.544 \text{ m}}{6{,}372{,}161.544 \text{ m} + 304.8006 \text{ m}} = 0.9999521691 \tag{7}$$

GoSurvey computes (via `geo::SeaLevelScaleFactor`, then `ResolveTraverseScaleFactor`'s Combined path):

Quantity	**Hand formula above**	**GoSurvey computed**
k_{sea}	0.9999521691	0.9999521691
k_{grid} (unchanged by elevation)	—	0.9999590531
$k_{\text{combined}} = k_{\text{grid}} \cdot k_{\text{sea}}$	0.9999112247	0.9999112242

The hand-computed k_{sea} matches GoSurvey's own figure to 9 decimal places, and the hand-computed k_{combined} matches GoSurvey's figure to 6 decimal places (the small residual is rounding in the hand conversion of R to meters above). k_{combined} differs from $k_{\text{grid-only}}$ exactly because sea-level reduction is now included — proving the "Combined" and "Grid only" choices genuinely produce two different, individually correct numbers rather than one option silently doing nothing. **Verified by:** `tests/TransformationTests.cpp`, test case "ResolveTraverseScaleFactor resolves Combined/GridOnly and reports why it can't", reusing the same AG9976 fixture as the pre-existing REQ-360 datasheet test.

5 Proof of the refusal path

A drawing with no coordinate system assigned (the default — "No Datum, No Projection") must not silently apply an arbitrary number. `ResolveTraverseScaleFactor` is proven to return a non-empty reason string and leave the factor at exactly 1.0 for a drawing with an empty `zoneCode` — the Traverse Editor shows this reason next to the checkbox and computes as if the option were off until the drawing's coordinate system is fixed.

6 Proof the default (off) path is unchanged

The same 1000.0000 ft, bearing 90° leg with the option left off (its default state) produces `computedHorizDist` = `computedGroundHorizDist` = `computedEasting` = `closurePerimeter` = 1000.0000 ft exactly — bit-for-bit what `ComputeTraverse` produced before this change, since $d_{\text{grid}} = d_{\text{ground}} \cdot 1.0 = d_{\text{ground}}$ whenever `applyScaleFactor` is false.

7 Automated test summary

File	Test case	What it proves
TraverseCalc / TraverseTests.cpp	Scale factor off reproduces unscaled coordinates	$k = 1$ default path is a no-op
TraverseCalc / TraverseTests.cpp	Scale factor on scales distance, coordinates, and perimeter	$d_{\text{grid}} = d_{\text{ground}} \cdot k$ propagates to ΔE, easting, perimeter
TraverseCalc / TraverseTests.cpp	Scale factor kind defaults to Combined	Safe default
CadCommands_Geo / TransformationTests.cpp	ResolveTraverseScaleFactor resolves Combined/GridOnly and reports why it can't	No-projection refusal; GridOnly matches NGS datasheet to 10^{-8}; Combined = GridOnly with sea level off; Combined $\neq$ GridOnly with sea level on

All four cases pass under both the `GoSurveyTests` and `GoSurveySnapTests` executables (Catch2, MSVC Release build). Source: PR #833, branch `feat/req401-traverse-scale-factor`, requirement `REQ-401` in `spec/requirements.md`.